Experiment No. 11

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Experiment Title

Use JavaScript timers (setTimeout(), setInterval()) and ES6 modules (import/export) to build a dynamic countdown or class scheduler with real-time updates and responsive layout.

Software / Tools Required

- Visual Studio Code
- Web browser
- HTML5
- CSS3
- JavaScript ES6 modules
- JavaScript timer functions: setTimeout() and setInterval()

Practical — Student Planner and Class Countdown

Aim

To develop a responsive class scheduler that displays a live clock and date, renders a list of classes, and provides a countdown timer with pause, reset, and next-class controls using JavaScript timers and ES6 modules.

Files Used

File	Purpose
index.html	Defines the page structure and loads the main JavaScript file as an ES module.
script.js	Updates the clock and countdown, renders class cards, and handles user interactions.
timer.js	Exports the class schedule, countdown duration, and time-formatting functions.
style.css	Styles the interface and provides responsive layouts for different screen sizes.

Program Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Planner | Class Countdown</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <main class="container">

        <header class="header">
            <div class="brand">
                <div class="brand-icon">□</div>
                <div>
                    <h1>Student Planner</h1>
                    <p>CLASS COUNTDOWN</p>
```

```
        </div>
    </div>

    <div class="header-right">
        <div class="header-clock">
            <span class="clock-dot"></span>
            <div>
                <p>LOCAL TIME</p>
                <h2 id="headerClock">00:00:00 AM</h2>
            </div>
        </div>

        <div class="date-badge">
            <span>▦</span>
            <span id="headerDate">Today's Schedule</span>
        </div>
    </div>
</header>

<section class="main-grid">

    <div class="countdown-panel">
        <div class="panel-heading">
            <div class="subject-illustration" id="subjectIllustration">&lt;/&gt;</div>

            <div class="subject-heading">
                <p class="eyebrow">CURRENT CLASS</p>
                <h2 id="subjectName">Web Development</h2>
                <p id="classRoom">□ Computer Lab · Room 204</p>
                <p id="classTime">◷ 10:00 AM – 11:00 AM</p>
            </div>

            <span id="classStatus" class="status upcoming">UPCOMING</span>
        </div>

        <div class="timer-panel">
            <p id="countdownLabel" class="timer-label">CLASS STARTS IN</p>

            <div class="countdown">
                <div class="time-box">
                    <span id="hours">00</span>
                    <small>HOURS</small>
                </div>

                <span class="separator">:</span>

                <div class="time-box">
                    <span id="minutes">00</span>
                    <small>MINUTES</small>
                </div>

                <span class="separator">:</span>

                <div class="time-box">
                    <span id="seconds">10</span>
                    <small>SECONDS</small>
                </div>
            </div>

            <div class="progress-track">
                <div id="progressBar" class="progress-bar"></div>
            </div>

            <p id="timerMessage" class="timer-message">
                Get ready! Your class will begin shortly.
            </p>

            <div class="timer-illustration">
                <span>✎</span>
                <span>✧</span>
                <span>▤</span>
            </div>
        </div>
```

```
                <div class="controls">
                    <button id="pauseButton" class="btn btn-primary">⏸ Pause</button>
                    <button id="resetButton" class="btn btn-secondary">↻ Reset</button>
                    <button id="nextButton" class="btn btn-outline">⏭ Next Class</button>
                </div>
            </div>

            <section class="schedule-panel">
                <div class="schedule-heading">
                    <div>
                        <p class="eyebrow">YOUR DAY AT A GLANCE</p>
                        <h2>Class Schedule</h2>
                    </div>
                    <span id="classCount" class="count-badge">4 Classes</span>
                </div>

                <div id="scheduleList" class="schedule-list"></div>

                <div class="schedule-tip">
                    <span>□</span>
                    <p>Select a class to view its countdown.</p>
                </div>
            </section>

        </section>

        <section class="bottom-grid">
            <div class="clock-panel">
                <div>
                    <p class="eyebrow">YOUR LOCAL TIME</p>
                    <h2 id="liveClock">00:00:00 AM</h2>
                    <p id="currentDate">Loading date...</p>
                </div>

                <div class="clock-illustration">
                    <div class="mini-paper">
                        <span>STUDY PLAN</span>
                        <p>✓ Read notes</p>
                        <p>✓ Attend class</p>
                        <p>✓ Revise topics</p>
                    </div>
                </div>
            </div>

            <div class="reminder-panel">
                <div class="quote-mark">“</div>
                <div>
                    <p class="eyebrow">A LITTLE REMINDER</p>
                    <h2>Small steps today lead to big achievements tomorrow.</h2>
                    <p class="reminder-caption">Keep learning, keep growing.</p>
                </div>
                <div class="plant-illustration">□</div>
            </div>
        </section>

        <footer>
            Built with JavaScript Timers <span>•</span> ES6 Modules
        </footer>

    </main>

    <div id="toast" class="toast"></div>

    <script type="module" src="script.js"></script>
</body>
</html>
```

script.js

```
import {
    classSchedule,
    countdownDuration,
    formatTime,
```

```
    formatCountdown
} from "./timer.js";

const liveClock = document.getElementById("liveClock");
const headerClock = document.getElementById("headerClock");
const currentDate = document.getElementById("currentDate");
const headerDate = document.getElementById("headerDate");
const subjectName = document.getElementById("subjectName");
const classRoom = document.getElementById("classRoom");
const classTime = document.getElementById("classTime");
const classStatus = document.getElementById("classStatus");
const subjectIllustration = document.getElementById("subjectIllustration");
const countdownLabel = document.getElementById("countdownLabel");
const timerMessage = document.getElementById("timerMessage");
const progressBar = document.getElementById("progressBar");
const scheduleList = document.getElementById("scheduleList");
const pauseButton = document.getElementById("pauseButton");
const resetButton = document.getElementById("resetButton");
const nextButton = document.getElementById("nextButton");
const toast = document.getElementById("toast");

let selectedIndex = 0;
let remainingSeconds = countdownDuration;
let paused = false;
let classStarted = false;
let intervalId = null;
let toastTimeout = null;

function updateClock() {
    const now = new Date();
    const actualTime = formatTime(now);

    liveClock.textContent = actualTime;
    headerClock.textContent = actualTime;

    currentDate.textContent = now.toLocaleDateString("en-IN", {
        weekday: "long",
        day: "numeric",
        month: "long",
        year: "numeric"
    });

    headerDate.textContent = now.toLocaleDateString("en-IN", {
        weekday: "long",
        day: "numeric",
        month: "short",
        year: "numeric"
    });
}

function updateDisplay() {
    const time = formatCountdown(remainingSeconds);

    document.getElementById("hours").textContent = time.hours;
    document.getElementById("minutes").textContent = time.minutes;
    document.getElementById("seconds").textContent = time.seconds;

    const progress =
        (remainingSeconds / countdownDuration) * 100;

    progressBar.style.width = progress + "%";
}

function showToast(message) {
    toast.textContent = message;
    toast.classList.add("show");

    clearTimeout(toastTimeout);

    toastTimeout = setTimeout(function () {
        toast.classList.remove("show");
    }, 3000);
}

function renderSelectedClass() {
```

```
    const item = classSchedule[selectedIndex];

    subjectName.textContent = item.subject;
    classRoom.textContent = "□ " + item.room;
    classTime.textContent = "◷ " + item.start + " – " + item.end;
    subjectIllustration.textContent = item.icon;

    classStatus.textContent = classStarted
        ? "CLASS IN SESSION"
        : "UPCOMING";

    classStatus.className = classStarted
        ? "status started"
        : "status upcoming";

    countdownLabel.textContent = classStarted
        ? "CLASS STATUS"
        : "CLASS STARTS IN";

    timerMessage.textContent = classStarted
        ? item.subject + " has begun. Happy learning!"
        : paused
            ? "Countdown paused. Resume when you're ready."
            : "Get ready! Your class will begin shortly.";

    pauseButton.disabled = classStarted;

    pauseButton.textContent = classStarted
        ? "Class Started"
        : paused
            ? "▶ Resume"
            : "⏸ Pause";

    renderSchedule();
}

function renderSchedule() {
    scheduleList.innerHTML = "";

    classSchedule.forEach(function (item, index) {
        const card = document.createElement("div");
        let statusText = "UPCOMING";

        if (index < selectedIndex) {
            statusText = "COMPLETED";
        } else if (index === selectedIndex && classStarted) {
            statusText = "IN SESSION";
        } else if (index === selectedIndex) {
            statusText = paused ? "PAUSED" : "SELECTED";
        }

        card.className = "schedule-item";

        if (index === selectedIndex) {
            card.classList.add("selected");
        }

        if (index < selectedIndex) {
            card.classList.add("finished");
        }

        card.innerHTML = `
            <div class="schedule-time">
                ${item.start}
                <br>
                ${item.end}
            </div>

            <div class="schedule-info">
                <h3>${item.subject}</h3>
                <p>□ ${item.room}</p>
                <span class="mini-status">${statusText}</span>
            </div>
```

```
        `;

        card.addEventListener("click", function () {
            selectClass(index);
        });

        scheduleList.appendChild(card);
    });

    document.getElementById("classCount").textContent =
        classSchedule.length + " Classes";
}

function selectClass(index) {
    clearInterval(intervalId);
    clearTimeout(toastTimeout);

    selectedIndex = index;
    remainingSeconds = countdownDuration;
    paused = false;
    classStarted = false;

    toast.classList.remove("show");

    document.querySelector(".countdown-panel").style.borderColor =
        "transparent";

    updateDisplay();
    renderSelectedClass();
    startCountdown();

    showToast(classSchedule[index].subject + " selected");
}

function beginClass() {
    clearInterval(intervalId);

    remainingSeconds = 0;
    classStarted = true;
    paused = false;

    updateDisplay();

    classStatus.textContent = "CLASS IN SESSION";
    classStatus.className = "status started";
    countdownLabel.textContent = "CLASS STATUS";

    timerMessage.textContent =
        classSchedule[selectedIndex].subject +
        " has begun. Happy learning!";

    pauseButton.disabled = true;
    pauseButton.textContent = "Class Started";

    document.querySelector(".countdown-panel").style.borderColor =
        "#b7dfbf";

    renderSchedule();

    showToast(
        "□ " + classSchedule[selectedIndex].subject + " has begun!"
    );
}

function startCountdown() {
    clearInterval(intervalId);

    intervalId = setInterval(function () {
        if (paused || classStarted) {
            return;
        }
```

```
        if (remainingSeconds > 0) {
            remainingSeconds--;
            updateDisplay();
        }

        if (remainingSeconds <= 0) {
            beginClass();
        }
    }, 1000);
}

pauseButton.addEventListener("click", function () {
    if (classStarted) {
        return;
    }

    paused = !paused;

    pauseButton.textContent = paused
        ? "▶ Resume"
        : "⏸ Pause";

    timerMessage.textContent = paused
        ? "Countdown paused. Resume when you're ready."
        : "Get ready! Your class will begin shortly.";

    renderSchedule();
});

resetButton.addEventListener("click", function () {
    clearInterval(intervalId);
    clearTimeout(toastTimeout);

    remainingSeconds = countdownDuration;
    paused = false;
    classStarted = false;

    document.querySelector(".countdown-panel").style.borderColor =
        "transparent";

    toast.classList.remove("show");

    updateDisplay();
    renderSelectedClass();
    startCountdown();

    showToast("Countdown reset");
});

nextButton.addEventListener("click", function () {
    const nextIndex = (selectedIndex + 1) % classSchedule.length;

    selectClass(nextIndex);
});

updateClock();
updateDisplay();
renderSelectedClass();

setInterval(updateClock, 1000);
startCountdown();
```

timer.js

```
export const classSchedule = [
    {
        subject: "Web Development",
        room: "Computer Lab · Room 204",
        start: "10:00 AM",
        end: "11:00 AM",
```

```
        icon: "</>"
    },
    {
        subject: "Data Structures",
        room: "Classroom · Room 302",
        start: "11:30 AM",
        end: "12:30 PM",
        icon: "{ }"
    },
    {
        subject: "Database Management",
        room: "Classroom · Room 105",
        start: "01:00 PM",
        end: "02:00 PM",
        icon: "⛁"
    },
    {
        subject: "Computer Networks",
        room: "Classroom · Room 210",
        start: "02:30 PM",
        end: "03:30 PM",
        icon: "⌘"
    }
];

export const countdownDuration = 10;

export function formatTime(date) {
    return date.toLocaleTimeString("en-IN", {
        hour: "2-digit",
        minute: "2-digit",
        second: "2-digit",
        hour12: true
    });
}

export function formatCountdown(totalSeconds) {
    const hours = Math.floor(totalSeconds / 3600);
    const minutes = Math.floor((totalSeconds % 3600) / 60);
    const seconds = totalSeconds % 60;

    return {
        hours: String(hours).padStart(2, "0"),
        minutes: String(minutes).padStart(2, "0"),
        seconds: String(seconds).padStart(2, "0")
    };
}
```

style.css

```
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

:root {
    --purple: #8062b4;
    --purple-dark: #45336d;
    --lavender: #f0e8ff;
    --green: #dff3e8;
    --green-dark: #39775f;
    --peach: #fff0e2;
    --text: #45385e;
    --muted: #8b819c;
    --border: #ece4f5;
}

body {
    font-family: Arial, sans-serif;
    background: linear-gradient(135deg, #fff3e5, #f6efff 55%, #e9f4f0);
    color: var(--text);
    min-height: 100vh;
    padding: 28px 18px;
}

.container {
    max-width: 1250px;
    margin: auto;
}

.header {
    background: #fff;
    border: 1px solid #eee5f6;
    border-radius: 22px;
    padding: 20px 26px;
    display: flex;
    align-items: center;
    justify-content: space-between;
```

```
    flex-wrap: wrap;
    gap: 18px;
    margin-bottom: 25px;
    box-shadow: 0 8px 30px rgba(75, 52, 104, 0.05);
}

.brand,
.header-right,
.header-clock {
    display: flex;
    align-items: center;
}

.brand {
    gap: 15px;
}

.brand-icon {
    width: 55px;
    height: 55px;
    background: var(--lavender);
    border-radius: 17px;
    display: grid;
    place-items: center;
    font-size: 29px;
}

.brand h1 {
    font-family: Georgia, serif;
    color: var(--purple-dark);
    font-size: 27px;
    margin-bottom: 5px;
}

.brand p {
    font-size: 10px;
    font-weight: bold;
    letter-spacing: 2px;
    color: var(--purple);
}

.header-right {
    gap: 14px;
    flex-wrap: wrap;
}

.header-clock {
    gap: 11px;
    background: #e5f2ed;
    border: 1px solid #d4e9df;
    border-radius: 13px;
    padding: 10px 15px;
}

.clock-dot {
    width: 9px;
    height: 9px;
    background: #4d9b75;
    border-radius: 50%;
    box-shadow: 0 0 0 4px #cce9d8;
}

.header-clock p {
    font-size: 9px;
    font-weight: bold;
    letter-spacing: 1.3px;
    color: #578b73;
    margin-bottom: 4px;
}

.header-clock h2 {
    font-size: 17px;
    color: #286a55;
    font-variant-numeric: tabular-nums;
}

.date-badge {
    display: flex;
    gap: 10px;
    align-items: center;
    background: var(--lavender);
    color: var(--purple-dark);
    border-radius: 12px;
    padding: 13px 16px;
    font-size: 12px;
    font-weight: bold;
}
```

```
.main-grid {
    display: grid;
    grid-template-columns: 1.15fr 0.85fr;
    gap: 22px;
    align-items: stretch;
}


.countdown-panel,
.schedule-panel {
    background: #fff;
    border: 1px solid var(--border);
    border-radius: 25px;
    padding: 28px;
    box-shadow: 0 10px 30px rgba(75, 52, 104, 0.05);
    min-width: 0;
}


.countdown-panel {
    background: linear-gradient(145deg, #fff, #f6efff);
    border: 2px solid transparent;
    transition: border-color 0.3s;
}


.panel-heading {
    display: grid;
    grid-template-columns: 70px minmax(0, 1fr);
    gap: 16px;
    align-items: center;
    position: relative;
    padding: 30px 0 25px;
}


.subject-illustration {
    width: 70px;
    height: 78px;
    border-radius: 20px;
    display: grid;
    place-items: center;
    background: #e7e2ff;
    color: #7255bd;
    border: 1px solid #d9d0ff;
    font-size: 22px;
    font-weight: bold;
}


.eyebrow {
    color: #9174c5;
    font-size: 10px;
    font-weight: bold;
    letter-spacing: 1.7px;
    margin-bottom: 9px;
}


.subject-heading h2,
.schedule-heading h2 {
    font-family: Georgia, serif;
    color: var(--purple-dark);
    font-size: 25px;
    margin-bottom: 9px;
}


.subject-heading > p:not(.eyebrow) {
    font-size: 12px;
    line-height: 1.8;
    color: var(--muted);
}


.status {
    position: absolute;
    right: 0;
    top: 0;
    padding: 8px 12px;
    border-radius: 20px;
    font-size: 9px;
    font-weight: bold;
    letter-spacing: 0.4px;
    white-space: nowrap;
}


.status.upcoming {
    background: var(--green);
    color: var(--green-dark);
}


.status.started {
    background: #d8f2dc;
    color: #28733a;
}
```

```
.status.completed {
    background: #efebf4;
    color: #81768e;
}


.timer-panel {
    background: #eee7ff;
    border: 1px solid #e2d5ff;
    border-radius: 22px;
    padding: 27px 18px 24px;
    text-align: center;
}


.timer-label {
    color: #8062b4;
    font-size: 11px;
    font-weight: bold;
    letter-spacing: 2px;
    margin-bottom: 25px;
}


.countdown {
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 9px;
}


.time-box {
    width: min(29%, 135px);
    min-width: 0;
    height: 108px;
    background: #faf7ff;
    border: 1px solid #dfd1fb;
    border-radius: 18px;
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    gap: 12px;
    box-shadow: 0 5px 14px rgba(88, 59, 135, 0.05);
}


.time-box span {
    font-size: clamp(27px, 3.5vw, 44px);
    font-weight: bold;
    color: #4f347f;
    font-variant-numeric: tabular-nums;
    line-height: 1;
}


.time-box small {
    font-size: 9px;
    color: #9382b2;
    font-weight: bold;
    letter-spacing: 1.1px;
}


.separator {
    color: #9279c1;
    font-size: 30px;
    font-weight: bold;
    margin-top: -14px;
}


.progress-track {
    height: 9px;
    background: #dcd0f5;
    border-radius: 20px;
    overflow: hidden;
    margin-top: 25px;
}


.progress-bar {
    width: 100%;
    height: 100%;
    background: linear-gradient(90deg, #8062b4, #a68be2);
    border-radius: 20px;
    transition: width 0.4s linear;
}


.timer-message {
    color: #817392;
    font-size: 12px;
    line-height: 1.6;
    margin-top: 18px;
    min-height: 20px;
}
```

```
.timer-illustration {
    display: flex;
    justify-content: center;
    gap: 25px;
    margin-top: 17px;
    color: #b29ad9;
    font-size: 20px;
}

.controls {
    display: grid;
    grid-template-columns: 1fr 1fr 1.2fr;
    gap: 10px;
    margin-top: 20px;
}

.btn {
    border-radius: 11px;
    padding: 14px 8px;
    font-size: 12px;
    font-weight: bold;
    cursor: pointer;
    transition: transform 0.2s, background 0.2s;
}

.btn:hover {
    transform: translateY(-2px);
}

.btn:disabled {
    opacity: 0.65;
    cursor: not-allowed;
    transform: none;
}

.btn-primary {
    border: 1px solid var(--purple);
    background: var(--purple);
    color: white;
}

.btn-primary:hover {
    background: #6d50a0;
}

.btn-secondary {
    border: 1px solid #f7ddc8;
    background: var(--peach);
    color: #a75e32;
}

.btn-secondary:hover {
    background: #ffe4cf;
}

.btn-outline {
    border: 1px solid #c9b4ee;
    background: white;
    color: #654a94;
}

.btn-outline:hover {
    background: #f6f0ff;
}

.schedule-panel {
    display: flex;
    flex-direction: column;
}

.schedule-heading {
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 12px;
    margin-bottom: 23px;
}

.schedule-heading h2 {
    font-size: 25px;
    margin-bottom: 0;
}

.count-badge {
    background: var(--green);
    color: var(--green-dark);
    padding: 9px 12px;
```

```
    border-radius: 20px;
    font-size: 10px;
    font-weight: bold;
    white-space: nowrap;
}

.schedule-list {
    display: flex;
    flex-direction: column;
    gap: 12px;
}

.schedule-item {
    display: grid;
    grid-template-columns: 62px minmax(0, 1fr);
    align-items: center;
    gap: 13px;
    border: 1px solid #eee8f5;
    background: white;
    border-radius: 15px;
    padding: 15px 13px;
    cursor: pointer;
    transition: background 0.2s, border-color 0.2s, transform 0.2s;
}

.schedule-item:hover {
    border-color: #c8b5eb;
    transform: translateY(-2px);
}

.schedule-item.selected {
    background: #f2ecff;
    border-color: #bca4e8;
}

.schedule-item.finished {
    opacity: 0.7;
}

.schedule-time {
    background: #f0e8ff;
    color: #7659a8;
    border-radius: 11px;
    padding: 11px 4px;
    text-align: center;
    font-size: 10px;
    font-weight: bold;
    line-height: 1.6;
}

.schedule-info {
    min-width: 0;
}

.schedule-info h3 {
    color: #51416e;
    font-size: 13px;
    line-height: 1.4;
    margin-bottom: 5px;
}

.schedule-info p {
    color: #978ca5;
    font-size: 10px;
    line-height: 1.5;
}

.mini-status {
    display: inline-block;
    margin-top: 7px;
    font-size: 9px;
    font-weight: bold;
    color: #6d5a8d;
    background: #f2eef7;
    padding: 5px 8px;
    border-radius: 10px;
}

.schedule-item.selected .mini-status {
    background: #dff3e8;
    color: #39775f;
}

.schedule-tip {
    display: flex;
    gap: 10px;
    align-items: center;
    background: #fff3e7;
```

```
    border-radius: 12px;
    padding: 13px;
    margin-top: 20px;
}

.schedule-tip span {
    font-size: 18px;
}

.schedule-tip p {
    color: #987657;
    font-size: 11px;
    line-height: 1.5;
}

.bottom-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 20px;
    margin-top: 22px;
}

.clock-panel,
.reminder-panel {
    border-radius: 22px;
    padding: 25px;
    min-height: 175px;
    position: relative;
    overflow: hidden;
}

.clock-panel {
    background: linear-gradient(120deg, #e4f6ec, #f2fbf5);
    border: 1px solid #d4eddf;
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.clock-panel .eyebrow {
    color: #578b73;
}

.clock-panel h2 {
    color: #286a55;
    font-size: clamp(25px, 3vw, 36px);
    margin: 12px 0 8px;
    font-variant-numeric: tabular-nums;
}

.clock-panel > div > p:last-child {
    font-size: 11px;
    color: #719181;
}

.clock-illustration {
    align-self: flex-end;
    transform: rotate(5deg);
}

.mini-paper {
    background: white;
    border: 2px solid #b8dec8;
    border-radius: 12px;
    padding: 12px;
    color: #56876f;
    box-shadow: 0 5px 12px rgba(71, 122, 90, 0.07);
}

.mini-paper span {
    display: block;
    background: #dff3e8;
    padding: 6px;
    text-align: center;
    border-radius: 5px;
    font-size: 8px;
    font-weight: bold;
    margin-bottom: 9px;
}

.mini-paper p {
    font-size: 9px;
    margin-top: 7px;
    white-space: nowrap;
}

.reminder-panel {
    background: linear-gradient(120deg, #fff1e6, #fffaf4);
```

```
    border: 1px solid #f5e1d0;
    display: flex;
    align-items: flex-start;
    gap: 14px;
}

.quote-mark {
    font-family: Georgia, serif;
    color: #d3a4d8;
    font-size: 55px;
    line-height: 0.8;
}

.reminder-panel .eyebrow {
    color: #b77d72;
    margin-top: 6px;
}

.reminder-panel h2 {
    font-family: Georgia, serif;
    color: #6d4a63;
    font-size: 20px;
    line-height: 1.5;
    max-width: 340px;
}

.reminder-caption {
    margin-top: 9px;
    color: #b18b81;
    font-size: 11px;
}

.plant-illustration {
    position: absolute;
    right: 16px;
    bottom: 8px;
    font-size: 37px;
}

footer {
    text-align: center;
    color: #9d90ad;
    font-size: 11px;
    margin: 25px 0 5px;
}

footer span {
    color: #bba7d8;
    margin: 0 5px;
}



.toast {
    position: fixed;
    left: 50%;
    bottom: 30px;
    z-index: 1000;


    min-width: 340px;
    max-width: calc(100% - 32px);
    padding: 20px 32px;


    background: #49366e;
    color: #ffffff;
    border: 1px solid #a58acb;
    border-radius: 16px;


    text-align: center;
    font-size: 15px;
    font-weight: 600;
    line-height: 1.6;


    box-shadow: 0 12px 35px rgba(73, 54, 110, 0.25);


    opacity: 0;
    transform: translate(-50%, 20px);
    pointer-events: none;


    transition: opacity 0.3s ease, transform 0.3s ease;
}

.toast.show {
    opacity: 1;
    transform: translate(-50%, 0);
}
```

```
@media (max-width: 480px) {
    .toast {
        min-width: 0;
        width: calc(100% - 32px);
        padding: 17px 20px;
        font-size: 13px;
        bottom: 18px;
    }
}



@media (max-width: 700px) {
    .header-right {
        width: 100%;
        justify-content: space-between;
    }
}

@media (max-width: 600px) {
    body {
        padding: 14px 10px;
    }

    .header {
        padding: 17px;
    }

    .brand-icon {
        width: 45px;
        height: 45px;
        font-size: 23px;
    }

    .brand h1 {
        font-size: 21px;
    }

    .date-badge {
        font-size: 10px;
        padding: 10px;
    }

    .countdown-panel,
    .schedule-panel {
        padding: 19px;
    }

    .panel-heading {
        grid-template-columns: 55px minmax(0, 1fr);
        gap: 12px;
    }

    .subject-illustration {
        width: 55px;
        height: 60px;
        font-size: 18px;
    }

    .subject-heading h2 {
        font-size: 21px;
    }

    .timer-panel {
        padding: 23px 9px;
    }

    .countdown {
        gap: 5px;
    }

    .time-box {
        height: 87px;
        border-radius: 12px;
        gap: 9px;
    }

    .time-box span {
        font-size: clamp(23px, 7vw, 34px);
    }

    .time-box small {
        font-size: 8px;
        letter-spacing: 0.6px;
    }
```

```
    .separator {
        font-size: 22px;
    }

    .controls {
        grid-template-columns: 1fr 1fr;
    }

    .btn-outline {
        grid-column: 1 / -1;
    }

    .bottom-grid {
        grid-template-columns: 1fr;
    }

    .schedule-heading h2 {
        font-size: 22px;
    }

    .header-clock h2 {
        font-size: 15px;
    }
}
```

Working / Concepts Used

Concept / Function	Purpose
setInterval()	Updates the countdown every second and refreshes the live clock.
setTimeout()	Hides the toast notification after three seconds.
clearInterval()	Stops the previous countdown interval before starting or resetting a timer.
clearTimeout()	Cancels a pending toast timeout when another notification or action occurs.
export	Makes the schedule data and helper functions available from timer.js.
import	Imports classSchedule, countdownDuration, formatTime, and formatCountdown into script.js.
Date()	Obtains the current date and time.
toLocaleTimeString()	Formats the current time for display.
DOM manipulation	Updates text, progress bar, status labels, and schedule cards dynamically.
Event listeners	Handle pause/resume, reset, next class, and selecting a class.
Responsive CSS	Uses grid layouts and media queries to adapt the interface to smaller screens.

Features

- Displays the local time and current date in real time.
- Shows the selected class, room, start time, end time, and status.
- Displays a countdown in hours, minutes, and seconds with a progress bar.
- Allows the user to pause or resume, reset the countdown, and switch to the next class.
- Allows a class to be selected from the schedule list.
- Displays a notification when a class is selected, the timer is reset, or a class begins.
- Uses a responsive lavender, green, and peach interface.

Output Screenshot

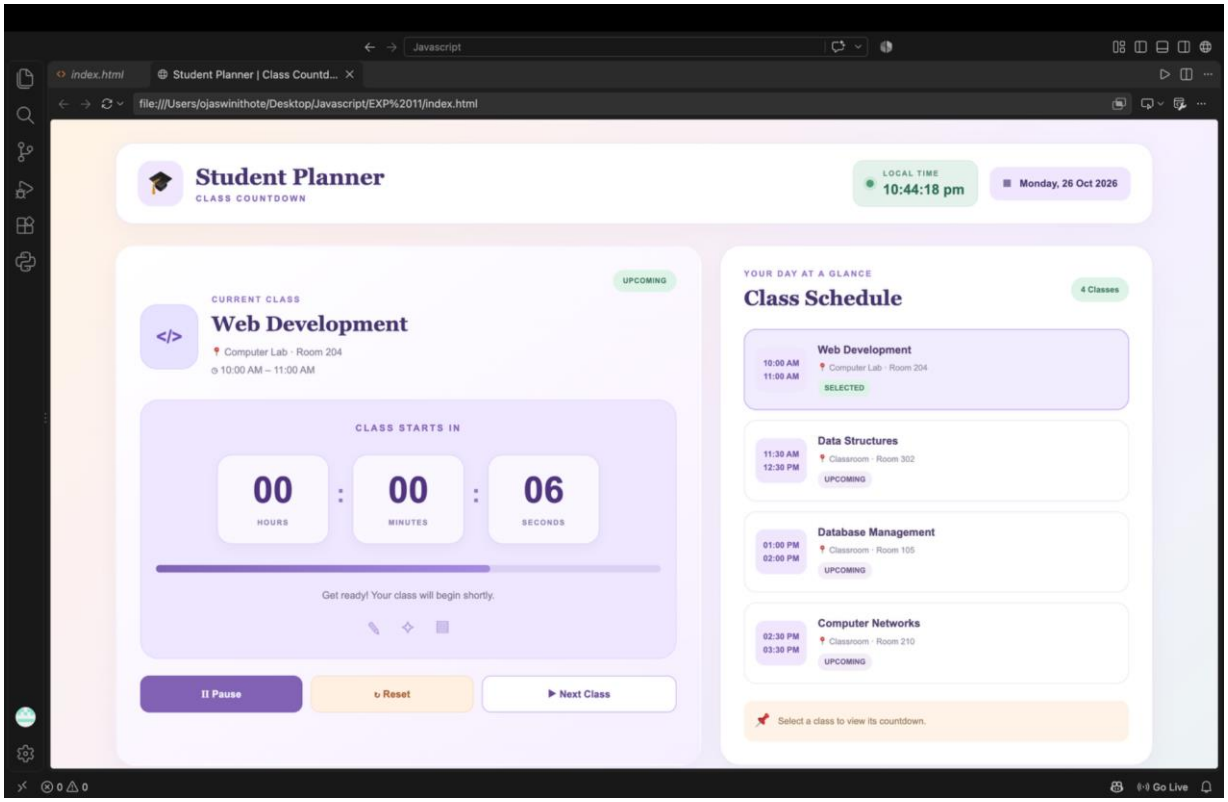


Figure 1: Student Planner and Class Countdown interface

Result / Conclusion

The Student Planner demonstrates how setInterval() and setTimeout() can be used for real-time updates and temporary notifications. ES6 import/export modules separate schedule data and time-formatting utilities from the interface logic. The final application dynamically displays class information, provides countdown controls, and uses responsive CSS to support different screen sizes.